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Gender Quotas in Hiring Committees: A Boon or a Bane for Women?

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Abstract. Women are underrepresented in many prestigious positions. Could increasing the share of women in hiring committees boost the rates at which women are hired into these positions? I use a difference-in-differences design to examine the effects of a French law on academic hiring committees that required each gender to represent a share of at least 40% of members. Contrary to the objectives of the law, I show that the reform backfired and significantly lowered women's probability of being hired. Because the negative effect of the reform is concentrated in committees headed by men, this result seems driven by the reaction of men to the reform. I find little evidence that the reform affects supply-side characteristics, such as the likelihood of women applying. The results suggest that the underrepresentation of women is unlikely to be solved by simply increasing the share of women in hiring committees or interview panels.

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1. Introduction

There is a large and persistent gender gap in the representation of women in the most prestigious positions. For instance, women in the United States were only 23% of C-suite executives and 9% of Fortune 500 Chief Executive Officers (CEOs) in 2021 (McKinsey 2022). There are many potential causes for this gender gap, such as the unequal distribution of household work and child-care (Goldin 2014, Blau and Kahn 2017), differences in attitudes to risk and competition (Niederle and Vesterlund 2007), discrimination against women (Llorens et al. 2021), or higher exposure to harassment (Folke and Rickne 2022). If discrimination at the hiring stage is one of the most salient causes (Moss-Racusin et al. 2012), especially in professions where women are underrepresented (Campero and Fernandez 2019), then gender quotas for the members of hiring committees or interview panels could be a solution to the underrepresentation of women. In theory, these quotas could not only reduce demand-side bias from men in the committees but also increase the number and quality of women applying for prestigious jobs by changing the beliefs that women have about the hiring process (Fernandez-Mateo and Fernandez 2016, Lepage et al. 2022).

Yet, we have little evidence on whether changing the gender composition of hiring committees through quotas actually helps women. Previous papers studying

this issue rely on random assignment of women to committees (Bagues et al. 2017), but quotas could generate different demand- and supply-side responses. Whether women prefer to hire other women remains unclear (Duguid et al. 2012, Faniko et al. 2021). Even if women prefer to work with other women (Byrne 1971), simply increasing their share in a committee might not lead to changes in hiring outcomes. Women could struggle to influence the committees' decisions (Radbruch and Schiprowski 2023), especially if the new members are less senior or qualified than the men they replace (Smith 2018), or fear backlash if they advocate for their preferred candidates (Hekman et al. 2017). Finally, men could also change their behavior when sitting in committees with a greater share of women.

This article fills the gap by examining the effect of a gender quota for hiring committee members on the hiring of women, taking advantage of a reform in France. To reduce gender bias in recruitment, the reform stated that from 2015 onward, each gender must be represented by a share of at least 40% of members in public sector academic hiring committees.¹ This feature makes it possible to estimate the causal effect of the reform by employing a difference-in-differences design. Using data collected by hand from three university archives, I compare fields in universities where hiring committees were on average above or below the 40% threshold prequota.

First, the quota has a large negative effect on the hiring of women of around 18 percentage points (p.p.), relative to a 43-p.p. counterfactual. Second, the quota has no significant effect on the gender composition of the applicant pool and other candidate characteristics, such as quality and academic connections. Finally, I study whether the reform changed other dimensions of the recruitment process, such as the share of female committee presidents. Although the share of women in recruitment committees moves from 40% to 47% in my sample, there is no increase in the share of women elected committee president, the key position in these committees. I find no change in the average quality of the committee, as proxied by publications, following the reform either.

What is the cause of this surprising negative effect of quotas on the hiring of women? Either women are less likely or unable to hire other women, or men change their behavior as a result of the reform. I find that the negative effect of the reform is entirely driven by committees headed by men. This suggests that changes in the behavior of men drive the negative effect of the reform.

When evaluating the impact of gender quotas, it is important to remember that they also come with significant costs for women. The relatively small number of women eligible to participate in committees implies that quotas increase their administrative workload, potentially harming their ability to publish. Given that women are more likely to volunteer for nonpromotable tasks than men, as shown in Babcock et al. (2017), increased participation in hiring committees could widen gaps in representation rather than reduce them.

This paper makes several contributions to the literature on gender inequality. Empirically, it provides new insights to the literature that examines the effect of the feminization of hiring committees. Previous papers that study the effect of greater gender diversity in hiring committees on the hiring of women rely on random assignment of women to committees rather than studying quotas (Bagues and Esteve-Volart 2010, Bagues et al. 2017). This paper is the first to directly estimate the effect of gender quotas in hiring committees on the recruitment of women and other demand- and supply-side outcomes. Theoretically, the paper suggests that the feminization of hiring committees can backfire if men change their voting behavior when more women are added to the committee (Akerlof and Kranton 2000, Monin and Miller 2001). The reaction of men, and of male leaders in particular (Stoddard et al. 2020), is crucial to understanding the outcomes from feminizing hiring committees (Bagues et al. 2017, Mocanu 2022). Consistent with previous work on diversity training (Kaiser et al. 2013, Dobbin and Kalev 2018), the results suggest that diversity initiatives with little managerial engagement can have unintended negative effects.

1.1. Related Literature

The continuing underrepresentation of women in the most prestigious positions has generated a large literature in management research (Fernandez-Mateo and Kaplan 2018, Azmat and Boring 2020). The reasons for this underrepresentation are multifaceted, and a large debate exists concerning the importance of discrimination at the hiring stage in explaining this gender gap (Moss-Racusin et al. 2012, Azmat and Petrongolo 2014). Recent audit studies find little evidence of gender bias at the hiring stage (Carlsson et al. 2021, Dares 2021, Birkelund et al. 2022), but the average null result could mask heterogeneity across occupations (Breda and Ly 2015, Breda and Hillion 2016, Campero and Fernandez 2019, Hangartner et al. 2021).

One policy solution to reduce discrimination at the hiring stage is to increase diversity in hiring committees. In general, the design of hiring practices has important consequences for gender disparities (Mocanu 2022). Quotas could be effective if marginalized individuals prefer to work with others they perceive to be similar to themselves (Byrne 1971). Alternative theoretical frameworks, such as value threat, predict instead that individuals with low categorical status are more likely to show out-group favoritism out of a desire to be considered valued members of the dominant group in the workplace (Duguid et al. 2012). Implicit bias research suggests that women hold gender biases as strong as those of men (Nosek et al. 2007). Empirical evidence on the own- or opposite-gender preferences of evaluators has yielded conflicting results. Some find that female evaluators have a positive effect on the outcomes for women (Lincoln et al. 2012, Boring 2017, Edo et al. 2019, Zeltzer 2020), whereas others find no effect (Abrevaya and Hamermesh 2012, Williams and Ceci 2015, Feld et al. 2016, Card et al. 2020). Broder (1993), Ellemers et al. (2004), and Hospido and Sanz (2021) find opposite-gender effects in grant applications, the evaluation of doctoral students' work ethic, and submissions to economic conference, respectively. It is, therefore, unclear *ex ante* whether we should expect women to be more positively inclined toward other women.

Allowing that women prefer to hire other women, quotas can only improve the hiring prospects of women if the new members are able to influence the behavior of the hiring committee. There is some evidence that increasing gender diversity has an effect on team behavior (Apesteguia et al. 2012, Azmat 2014), although women may have less influence than men in group decisions (Radbruch and Schiprowski 2023). Previous papers have found that board quotas modify firm outcomes (Matsa and Miller 2013, Maghin 2022), for instance. However, even if a change in gender diversity affects decision making, the effect might not necessarily carry over when it comes to advocating for more gender diversity. Women advocating for more

gender equality could face issue legitimacy and struggle to convince others of the business importance of equality (Hussain et al. 2023), or they might fear backlash from men (Hekman et al. 2017). Papers studying the “trickle-down” benefits of boardroom quotas find that they have a positive effect on the wages of women in the boardroom but that these gains do not trickle down the wage ladder (Bertrand et al. 2019, Dalvit et al. 2022, Maida and Weber 2022). More generally, although these boards have the power to choose CEOs, there has been no commensurate increase in the share of female CEOs in countries that have passed gender quotas (Deloitte 2021).

A final issue to consider is that the behavior of men might also change as a consequence of the reform. Being exposed to more female peers or having more women in group decision making has been shown to improve the attitudes of men toward other women (Battaglini et al. 2020, Stoddard et al. 2020, Dahl et al. 2021, Born et al. 2022). On the other hand, some empirical and theoretical papers suggest that men may react negatively as a result of the quota. The moral licensing effect (Monin and Miller 2001, Blanken et al. 2015), for instance, proposes that previous good actions, such as accepting a gender quota, can act as a justification for future immoral acts, such as refusing to hire women. Akerlof and Kranton (2000) posit that individuals experience discomfort when faced with threats to their identity. The quota could be considered as a threat to male identity, generating a backlash effect (Leibbrandt et al. 2018, Ip et al. 2020). A final possibility is that men expect women to be biased in favor of other women and adjust their votes strategically as a result (Snellman and Solal 2023).

A small body of research has studied the effect of the feminization of committees on women. This literature has mainly inferred the effects of gender quotas on the basis of the random assignment of women to committees. Bagues and Esteve-Volart (2010) and Bagues et al. (2017) find that women are less likely to be hired when more women are randomly assigned to a candidate's promotion committee, with the second paper attributing this effect to the reaction of men.² The reform studied in this paper gives a direct estimate of the effect of gender quotas. This is important because quotas could potentially generate different demand-side (Li 2017) and supply-side (Solal and Snellman 2019) responses relative to random assignment or voluntary implementation. For instance, women chosen to participate in a hiring committee after the quota will be differently selected than before (Smith 2018, Bertrand et al. 2019), potentially leading to different hiring outcomes.

Previous articles have emphasized the difficulty of disentangling supply-side and demand-side effects in the hiring process (Fernandez-Mateo and Kaplan 2018). Characteristics that seem determined by supply-side

preferences, such as the gender composition and quality of the applicant pool, are also affected by demand-side factors, such as the work-life balance of the job (Barbulescu and Bidwell 2013) or expectations of discrimination (Pager and Pedulla 2015). For instance, women are less likely to apply for jobs when they believe the selection process is biased (Fernandez-Mateo and Fernandez 2016), and companies with diverse boards are more attractive to job applicants (Schäpers et al. 2021, 2023), which makes analyzing demand- or supply-side characteristics in isolation a perilous exercise.

Solutions have been to study cases where either the gender of candidates is invisible (Goldin and Rouse 2000) or where the candidates have no choice in deciding whether to apply (Fernandez-Mateo and Fernandez 2016). Studying an exogenous change in the feminization of hiring committees allows for a different approach to this problem because we can observe both the changes in demand and the reaction on the supply side. There are reasons to think that a quota could modify supply-side characteristics. Because women are more likely to be connected to other women (Boschini and Sjögren 2007), diversifying the pool of potential committee members could also increase the probability of being connected to a committee member. More women could decide to apply for positions if the quota durably shifts expectations on the organizational culture of academia. Given that quotas signal a shift in the preferences of policy makers and not the organization, such a shift could be smaller, however, than when the composition of hiring committees is changed voluntarily (Solal and Snellman 2019, Baron et al. 2021).

Many organizations are now facing recommendations to have more diverse interview panels, whether in government hiring³ or in the private sector,⁴ even though the empirical and theoretical evidence that they improve hiring outcomes is limited. Previous research has stressed that policies aiming to improve diversity can have unintended consequences (Dobbin et al. 2015, Dobbin and Kalev 2018). According to self-determination theory, which states that individuals react differently to intrinsic and extrinsic motivation (Deci and Ryan 2004), constraints on individual autonomy can also provoke resistance. Because quotas are externally imposed, they may lower the motivation of committee members to pursue diversity, as has been shown by Legault et al. (2011) and Chang et al. (2019b) in the case of diversity training. Moreover, by creating the illusion of a procedurally fair environment, such policies can ironically limit the ability to detect and combat discrimination (Kaiser et al. 2013, Brady et al. 2015). These unintended consequences should incite managers and policy makers to be wary when promoting new diversity initiatives and to study which policies have a track record of success.

In Section 2, I explain how I construct the data set, and present some descriptive statistics and background information. In Section 3, I look at how the quota affected hiring, and in Section 4, I look at how the quota affected supply-side and other committee characteristics. I then discuss the interpretation of these results and potential channels in Section 5.

2. Background and Data

In this section, I present the data set compiled from administrative data on 455 hiring committees from three different French universities.

2.1. French Hiring Committees

French academic hiring committees are created ad hoc for each position to be filled. These committees have a president, who has broad powers over the nomination of committee members. This committee president is chosen directly by the academic committee of the university and is quite often the head of the hiring department. The president also has a deciding vote in case of a tie between candidates. There is some variation in the number of members in these committees; by law, there must be between 8 and 20 committee members. At least half of the members must be from outside the hiring university.

Once a committee has been created, candidates can apply via a web platform called GALAXIE and post their Curriculum Vitae (CVs). In some (but not all) cases, they are aware of the committee composition at the time of their application. The committee then decides which candidates to invite to an on-campus job interview, and after these visits, it ranks the candidates

it deems worthy (if any) of hiring. The committee votes on the final ranking and not on the candidates individually. Although members can dissent during the final vote, the final ranking is almost always accepted unanimously. Because the individual behavior of committee members is not observed, I can infer the effect of increasing the number of women in committees from only two variables: the gender of the committee president and the percentage of women on the hiring committee.

Offers are made sequentially in the order in which candidates are ranked. All ranked candidates can potentially receive an offer from the university if the candidates ranked above them refuse the offer. This is relatively infrequent in the data set because only 20 first-ranked candidates refuse the position. The candidates are aware of the ranks that they receive, which helps to inform their decision on which offers they should accept or reject.

2.2. Descriptive Statistics

For each committee, I have access to the names and ages of the committee members and the names and ranks of the candidates as well as the gender composition of the candidate pool. For a small number of committees (72), I am missing data on applicants and use instead the percentage of women among the ranked candidates. The data span from 2009 to 2018, but the bulk of the data is from committees post-2013.

I present some statistics on the committees in Table 1. The number of female presidents and the share of women on committees vary strongly between fields. In general, the percentage of women hired is correlated with the percentage of female presidents and the share

Table 1. Descriptive Statistics on Recruitment

	No. of recruitments	No. of recruitment prequota	No. of recruitment postquota	Share of women hired	Share of female presidents	Share of female committee members	Share of female applicants
Biology	65	28	37	0.51	0.38	0.48	0.54
Business	18	4	14	0.33	0.22	0.49	0.42
Chemistry	18	5	13	0.39	0.44	0.44	0.39
Economics	27	11	16	0.19	0.07	0.41	0.26
Education	18	4	14	0.50	0.17	0.46	0.63
Engineering	18	5	13	0.06	0.11	0.40	0.23
History	12	10	2	0.33	0.42	0.37	0.28
Languages	40	16	24	0.47	0.47	0.52	0.56
Law	33	14	19	0.42	0.09	0.43	0.46
Maths	57	16	41	0.14	0.30	0.38	0.19
Pharmacology	30	9	21	0.57	0.40	0.48	0.53
Physical education	13	7	6	0.15	0.08	0.46	0.15
Physics	28	12	16	0.21	0.14	0.40	0.20
Political science	19	12	7	0.26	0.32	0.38	0.35
Psychology	30	14	16	0.57	0.43	0.50	0.62
Sociology	29	21	8	0.45	0.40	0.37	0.43
Total	455	188	267	0.36	0.30	0.44	0.40

Note. The statistics are compiled on a data set that includes assistant and full professors over the years 2009–2018.

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of women in committees at the field level but also with the percentage of female applicants.

During the analysis, the ranking variable includes the K -ranked candidates and the candidates who are interviewed but are not ranked, all of whom are given the rank $K + 1$. Because I use only order and not distance in my regressions, this normalization is harmless. Candidates who are not interviewed are discarded from the regression analysis but are included when I compute the gender ratio of applicants.

Although I do not observe the CVs of candidates, I can control for two of the most important components of candidate quality: publications and connections. To control for publications, I collect the h index and citations of candidates using the Scholar h -index calculator (Ianni 2010), which creates h indices and citation counts from Google Scholar. Citations were collected in 2018 for articles published by the year in which the committee convened.⁵ Academic connections are another potential confounder (Combes et al. 2008). To obtain more information on connections between committee members and candidates, I scraped the French website theses.fr, which gathers information on which institution French doctors received their PhD from and the name of their PhD supervisor.

2.3. The Reform

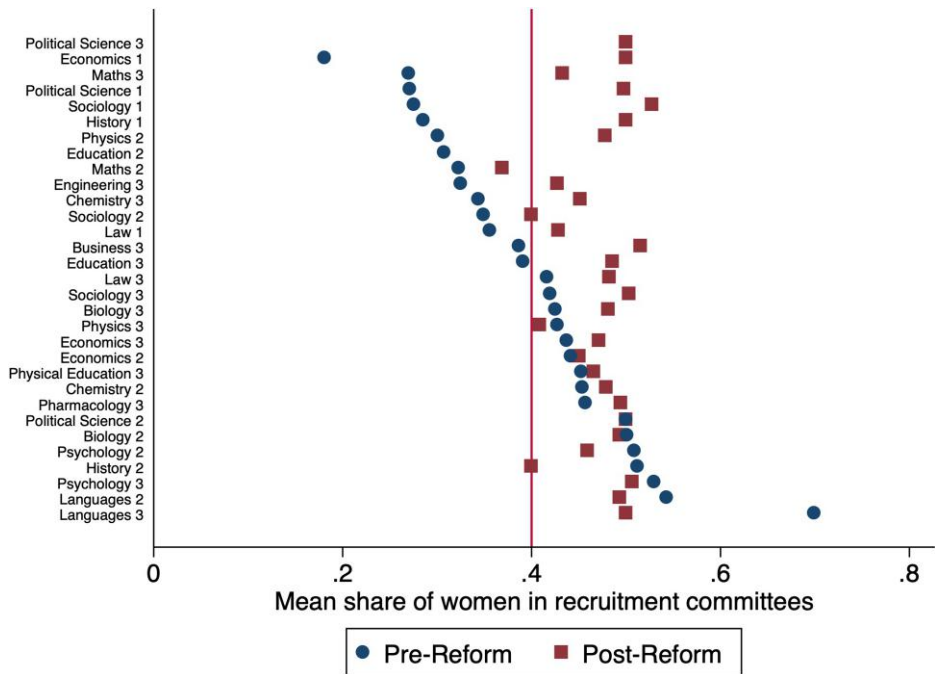
As part of the French government’s push for greater gender equality, since January 1, 2015, each committee

must have member shares of at least 40% for each gender. However, there are neither constraints on the gender of the committee president nor quotas on the gender of the candidates hired.

Although the objective of the law was to reduce gender discrimination, as the heading of the law explicitly states, it was not specifically targeted at academia. The reform was part of a general law that imposed gender parity more widely for French public servants, such as quotas for management positions within the French public administration.

Figure 1 shows how the reform affected the gender composition of hiring committees at the field-university level.⁶ I use variation at the field-university level to identify the effect of the quota, comparing fields in universities where the prereform average proportion of women in committees was above or below 0.4. All observations where the circle is to the left of the 0.4 threshold, represented by the vertical line, are considered to be treated in the main part of the analysis, whereas the others are considered to be untreated and are used as a comparison group. The treated departments are those that did not respect the quota *ex ante*. As is clear from the figure, the reform had a significant effect on this group; the mean share of women in committees moved from 30% to 45%, with a t statistic of 9.89. There was no significant effect of the quota for the comparison group, where the mean share moved from 0.480 to 0.482 (t statistic = 0.2). Also striking is that the

Figure 1. (Color online) Effect of the Quota on the Share of Women in Recruitment Committees



Note. Each entry represents a field (e.g., maths, law, biology) in a university (1, 2, 3).

average female representation in many cases shifted to 50%, even in committees with few women beforehand, suggesting a shift to a new social norm in favor of representation higher than the 40% threshold, as in Chang et al. (2019a).

3. Effect of the Quota on the Recruitment of Women

In this section, I present the effect of the quota on demand-side behavior. In Section 3.1, I estimate the effect of the reform on whether women are more likely to be hired. In Section 3.2, I estimate the effect of the quota on the ranks of women and discuss threats to validity in Section 3.3. In Section 4, I study the effect of the reform on applicant and committee characteristics.

3.1. Effect of the Quota on Hiring

I use the reform to compare two different groups: fields in universities where the gender composition of the committee was below or above 40% before 2015. For instance, if, on average, committees in law had a 35% share of women on average in University A and a 45% share on average in University B before 2015, then committees in law in University A are assigned to the treated group, and law committees in University B are considered to be untreated. In practice, most fields are assigned consistently across universities.

We can first consider whether the reform had any effect on first-ranked candidates. Because most candidates who are ranked first accept the offer, this tests directly for whether more women are hired when there are more women on the committee. I estimate the equation at the committee level to see whether women are more or less likely to be first ranked in the impacted group after the reform:

$$W_j = \alpha R_j T_j + \gamma T_j + \delta R_j + \beta_1 X_j + \epsilon_j,$$

where W_j is equal to one if the first-ranked candidate for a hiring committee j is a woman, T_j is an indicator variable equal to one if the prequota average share of women in committees at the field-university level is below the 40% threshold, R_j is an indicator variable equal to one if the committee takes place after 2015, and X_j is a vector of control variables such as field fixed effects and the share of female candidates. As in a standard difference-in-differences framework, γ represents the effect of being in the impacted group before the reform, δ is the effect of the reform on the comparison group, and α is the parameter of interest: the effect of the quota. This equation⁷ is estimated through ordinary least squares (OLS).⁸

The results are presented in columns (1), (2), and (3) of Table 2. Column (1) includes university and field fixed effects, whereas column (2) adds year and university $\times$ field fixed effects. I drop contests where all

the candidates are of the same gender in column (3). Demand-side bias should have no effect on these committees because the gender of the hired candidate is fully determined by the gender composition of the applicant pool. This column gives us an idea of the magnitude of the effect on contestable committees.⁹

Our coefficient of interest, the effect of being in the treated group postquota, can be found in the first row of Table 2 and shows a significant negative effect on the probability of a woman being hired. The point estimate suggests a drop in the number of women hired of 18 percentage points in committees affected by the quota and moves to 25 percentage points when excluding contests with applicants from a specific gender only. Looking at the other coefficients from Table 2, a slightly surprising result that is also found in the subsequent tables is that fields that were not feminized prereform actually tended to favor women, as can be seen from the coefficient in the third row. Then, in fields affected by the reform, women were less likely to be hired, whereas fields unaffected by the reform increased their hiring of women. This is surprising because one of the justifications for imposing a quota is that women may face discrimination in committees dominated by men, but this is not what we find prereform. I discuss whether this is credible in Section 3.3.

To partially control for differences in publications in this setting, I rank all the candidates by their h index. I then create a dummy variable equal to one if the candidate with the highest h index is a woman. This variable has a large and significant effect on the probability that a woman is hired. I cannot, however, control for other individual-specific characteristics, which may affect the magnitude of the results, such as academic connections, in this specification. This could bias the results if women are more likely to be connected to a committee member after the reform (Boschini and Sjögren 2007).

A solution to this problem is to estimate the hiring probability through a conditional logit.¹⁰ A conditional logit estimates the probability that candidate i is picked out of a pool of interviewed candidates, and it allows us to investigate the effect of individual-level characteristics on the probability of being hired.

I estimate the following specification:

$$F_{ij} = \alpha R_j T_j g_i + \gamma T_j g_i + \delta R_j g_i + \beta_1 X_i + \beta_2 g_i X_j + \epsilon_{ij},$$

where F_{ij} is a dummy variable equal to one if candidate i is ranked first, g_i is a binary variable equal to one if candidate i is a woman, X_i is a vector of individual characteristics of the candidate (such as the candidate's age-adjusted h index; the publication measure that is the most significant when all other measures are included, standardized by field and seniority; in some specifications, the candidate's gender; and whether the candidates received their PhD from the university to which they are applying), and X_j a vector of committee

Table 2. Effect of the Quota on the Gender of the First-Ranked Candidate

	Dependent variable				
	<i>First-ranked candidate is female</i>			<i>Ranked first</i>	
	Estimator: OLS			Estimator: Conditional logit	
	(1)	(2)	(3)	(4)	(5)
Interacted with gender of the candidate for conditional logit					
Postquota × below 40% threshold	−0.163** (0.07) [0.036]	−0.180** (0.08) [0.027]	−0.247** (0.11) [0.037]	−1.470** (0.71)	−1.733* (0.89)
Postquota	0.108** (0.05)	—	—	0.713** (0.36)	—
Below 40% threshold	0.147* (0.08)	—	—	1.022*** (0.39)	—
Female president	−0.045 (0.03)	−0.067 (0.04)	−0.072 (0.05)	−0.600** (0.29)	−0.645* (0.36)
Share of female candidates	0.659*** (0.06)	0.664*** (0.06)	0.688*** (0.15)	0.561 (1.30)	1.204 (1.42)
Professorship	0.129*** (0.04)	0.111*** (0.04)	0.124** (0.05)	0.971*** (0.37)	0.776** (0.36)
At candidate level for conditional logit					
Max <i>h</i> index of contest is female	0.335*** (0.04)	0.331*** (0.05)	0.334*** (0.05)	—	—
Standardized age-adjusted <i>h</i> index	—	—	—	0.356*** (0.06)	0.343*** (0.06)
PhD supervisor in the committee	—	—	—	0.452* (0.27)	0.387 (0.27)
PhD from the same university	—	—	—	0.583*** (0.17)	0.611*** (0.17)
Marginal effect of treatment	—	—	—	−0.152	−0.176
Mean of dependent variable	0.36	0.36	0.40	—	—
Interacted with gender for conditional logit					
Field fixed effects	Yes	No	No	Yes	No
University fixed effects	Yes	No	No	Yes	No
University × field fixed effects	No	Yes	Yes	No	Yes
Year fixed effects	No	Yes	Yes	No	Yes
Observations	455	455	359	1,357	1,357

Notes. Standard errors are clustered at the university × field level and can be found in parentheses. Wild-cluster bootstrapped *p*-values can be found in square brackets. The gender variable is coded as one for a woman and zero for a man. The treated group is constructed based on whether the average share of women in committees at the university-field level was below the 40% threshold prequota. Variables interacted with the gender of the candidate should be interpreted as the differential effect of the variable for women with respect to men. Contests with applicants only of a specific gender are dropped in column (3). Contests where only one candidate was interviewed or where only applicants of a specific gender applied are dropped from columns (4) and (5). The max *h*-index variable is a dummy variable equal to one if the maximum *h*-index of all candidates corresponds to a woman. Marginal effects represent the average change in the probability of being ranked first for women in the treated group postquota.

p* < 0.10; *p* < 0.05; ****p* < 0.01.

characteristics (such as the share of female applicants or a dummy for whether the post is for an assistant or associate professorship)¹¹ that are interacted with g_i .

Because the conditional logit model only exploits within-committee variation, we cannot directly estimate the effect of variables that do not vary within committee. The *h*-index variable, for instance, varies across candidates being evaluated by the same committee, which means that the effect of having better publications can be estimated for all candidates. Directly estimating the effect of being in one of the two groups, on

the other hand, is impossible because all candidates in a particular committee will either be affected or be unaffected by the quota. Hence, we need to interact all variables that are constant within committee with g_i , a binary variable equal to one if the candidate is a woman. The coefficients can then be interpreted as the differential effect of the variable for women relative to that for men.

The results are presented in columns (4) and (5) of Table 2. We can see that the reform has a large, significant, and negative effect on the hiring of women.¹² This effect

is robust (at the 5.1% level) to the use of university $\times$ field and year fixed effects interacted with the gender of the candidate. To interpret the size of the effect of the quota, one can compare the coefficient in the first row with the one for the standardized h index.¹³ In column (1), for instance, the size of the effect is equivalent to a three-standard deviation drop in the h index of candidates. To provide a more concrete estimate of how much the reform affected the hiring of women, I show the average change in the probability of being ranked first for women in the treated group postquota under the heading marginal effect of treatment.¹⁴ These estimates imply that individual women in the sample see their probability of being hired drop by 15–18 p.p.

The other control variables have the expected sign; having a supervisor in the committee or having a PhD from the university where you are applying improves the candidate's ranking by the committee, and the effect size here is larger than that of a one-standard deviation increase in the h index. This result confirms previous research on the importance of networks and connections in academia, such as Combes et al. (2008) or Colussi (2018).

3.2. Effect of the Quota on the Ranks of Candidates

Next, I take advantage of the full ranking data by using a rank-ordered logit. This method is similar to the conditional logit, but where the conditional logit only considers the hiring decision, the rank-ordered logit exploits variation across the entire ranking. For instance, a committee deciding to rank a particular candidate second or third will affect the estimated coefficients in a rank-ordered logit but leave the results from the conditional logit unchanged.

The rank-ordered logit is more precise because it exploits all the variation in the data,¹⁵ but the results could be affected by candidates lower down the rankings, which are inconsequential for hiring outcomes. Ranks could still be an important margin to consider, however, because women are more likely to leave academia than men (Geuna and Shibayama 2015) and are less likely to reapply for jobs once rejected (Fernandez-Mateo et al. 2023). By providing a signal that they are close to the bar, higher ranks could incite more women to stay in the academic job market.

I estimate the equation:

$$Rank_{ij} = \alpha R_j T_j g_i + \gamma T_j g_i + \delta R_j g_i + \beta_1 X_i + \beta_2 g_i X_j + \epsilon_{ij},$$

where $Rank_{ij}$ is the rank of the candidate, and the other variables are defined as before.

The results are presented in Table 3. For all variables that do not vary within committee, the results presented are the differential effects for women with respect to men.

Column (1) includes no control variables except for the h index and university fixed effects. I add gender $\times$ university $\times$ field and gender $\times$ year fixed effects in columns (2), (3), and (4). To investigate how labor supply and demand effects interact, I first show the results without controlling for any of the other control variables in column (2); then, I add the h index in column (3) and the other control variables in column (4). This should allow me to disentangle how the effect of the reform depends on changes in demand (committee composition) and changes in supply (for instance, better female applicants).

The coefficients in the first row of Table 3 give us the effects of the quota on the ranks of women, which are large and negative. Because the coefficients are relatively similar across columns, most of the effect seems driven by demand-side changes. The marginal effect, estimated in the same way as the conditional logit, suggests a drop in the probability of being ranked first between 8 and 10 p.p.

3.3. Threats to Validity

In this subsection, I discuss three potential threats to the validity of the difference in differences.

First, fields with few women prereform tended to favor women, but this effect disappears and even switches sign once the quota is put in place. This is clearly seen from Figure 2, which plots the year-on-year estimates for the two groups separately using the rank-ordered logit. One possible explanation for this pattern is mean reversion, which makes it important to consider whether it is likely that committees dominated by men were likely to favor women before the quota. In fact, this pattern is close to those found in Breda and Hillion (2016), which showed that women were favored in oral examinations for posts as secondary school teachers in fields that had few women instructors. One particularly interesting feature of this paper is that large shares of the graders are also university professors (80% for the hardest examination) and so, could also be part of the hiring committees discussed in this article. Their sample covers the period 2006–2013, overlapping in part with my sample. Given the evidence from this paper, our prior should be that women are favored in male-dominated fields, and so, the results shown here are unlikely to be because of mean reversion.

Second, the validity of the difference-in-differences estimates could be threatened by the presence of pre-trends in the data, especially because fields that select few women might be very different from those that select many. Visual inspection of Figure 2 shows no significant trends in either group before the reform. Furthermore, I formally test the significance of gender $\times$ year $\times$ below 40% threshold fixed effects in Table

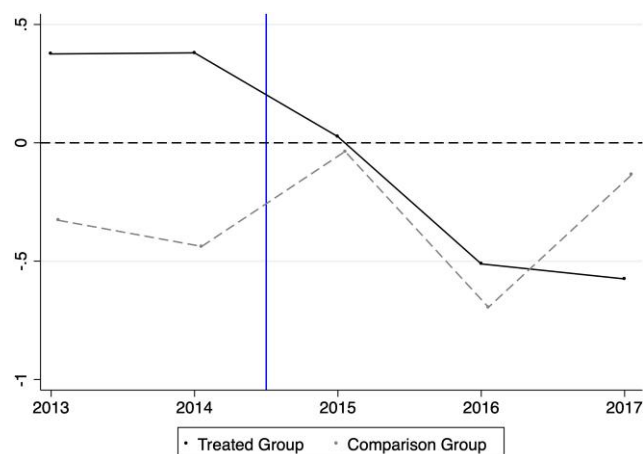
Table 3. Effect of the Quota on the Rank of Female Candidates

	Dependent variable: Rank			
	Estimator: Rank-ordered logit			
	(1)	(2)	(3)	(4)
Interacted with gender of the candidate				
Postquota × below 40% threshold	−0.952*** (0.36)	−0.696* (0.36)	−0.694** (0.35)	−0.894** (0.36)
Postquota	0.230 (0.28)	—	—	—
Below 40% threshold	0.689** (0.27)	—	—	—
Female president	—	—	—	−0.162 (0.19)
Variables at the candidate level				
Standardized age-adjusted <i>h</i> index	—	—	0.288*** (0.03)	0.290*** (0.03)
Marginal effect of treatment	−0.109	−0.0795	−0.0788	−0.102
Controls	None	None	<i>h</i> index	All
Gender × university fixed effects	Yes	No	No	No
Gender × university × field fixed effects	No	Yes	Yes	Yes
Gender × year fixed effects	No	Yes	Yes	Yes
Observations	1,357	1,357	1,357	1,357

Notes. Standard errors are clustered at the university × field level. The gender variable is coded as one for a woman and zero for a man. The treated group is constructed based on whether the average share of women in committees at the university-field level was below the 40% threshold prequota. Controls include the share of female candidates, president gender, professorship level, and academic connections. Contests where only one candidate was interviewed or where applicants of a specific gender applied are dropped from the analysis. Marginal effects represent the average change in the probability of being ranked first for women in the treated group postquota.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

OA.25 in the online appendix and find that none of the years prereform are significantly different from zero, which supports the idea that there are no differential pretrends.

Figure 2. (Color online) Effect of Gender on Ranks in the Treated and Comparison Groups

Notes. Dots represent coefficients from year dummies interacted with treated or comparison group status and the gender of the candidate. Estimates control for the standardized age-adjusted *h* index. The standard errors on each point are large and are omitted from the figure for clarity.

Third, the reform has a positive and significant effect on the comparison group in the specifications of Table 2. Although this could simply reflect an underlying improvement in the likelihood of women being hired in academia more generally, one might be worried that this is caused by contamination of the comparison group. However, there is little change in the average gender composition of the recruitment committees in this group (from 0.480 to 0.482). Furthermore, as argued and shown in Section 6, the rank-ordered logit estimates are more precise, and they show no significant effect of the reform on the comparison group. Finally, estimating the effect of the reform on the treated group alone or estimating a single constant fixed effect for the comparison group shows a significant negative effect of the reform, which implies that the effects estimated here are not driven by unrelated changes in the comparison group.

Three further robustness checks are discussed in Appendix A. Appendices A.1 and A.2 study the effect of the reform under different time frames and rank restrictions. Appendix A.3 then analyses the effect of the reform using a continuous specification as in Bertrand et al. (2019). Considering two years on either side of the reform only to rule out long-term trends, dropping lower-ranked candidates, or using a continuous specification gives the same qualitative results as in the previous tables.

4. Effect of the Quota on Supply-Side Characteristics and the Selection of Committee Members

In this section, I analyze the effect of the quota on dimensions other than the recruitment of women. Section 4.1 looks at the effect of the reform on supply-side characteristics, and Section 4.2 analyzes the effect of the reform on committee characteristics.

4.1. Effect of the Quota on Supply-Side Characteristics

A potential effect of the reform is to encourage more women to apply for academic positions. I test whether the gender composition of the applicants is affected by the reform in the first two columns of Table 4.

I estimate the following model using OLS:

$$ShW_j = \alpha R_j T_j + \beta_1 X_j + \phi_j + \psi_j + \epsilon_j,$$

where ShW_j represents the share of female applicants and where ϕ_j and ψ_j are year and university $\times$ field fixed effects, respectively. Because the publications of the candidate should not matter in this analysis, I only control for whether the position is for a professorship and the gender of the jury president. We are once again interested in the significance of α , which here is the effect of the quota on the gender composition of the applicant pool, and it is shown in the first row of column (1). There is no significant effect of the quota, but

the point estimate is positive and relatively large. There are two reasons for this. The first is that the positive point estimate is driven entirely by full professor applications, where the number of applicants is very small. The point estimate translates to an increase of less than half a woman per committee. The second cause is the long time span covered by my data, which feature committees from 2009 to 2018.

One way to investigate this problem is to restrict the time span to the period immediately around the reform, from 2013 to 2016, where the effect on the reform is strongly negative (see Table A.1). In this time frame, column (2) shows that the point estimate for the effect of the quota on the applicant pool is negative. This suggests that the changes in the applicant pool seen in column (1) could be because of longer-run trends in the propensity of women to apply for these positions rather than to the reaction of candidates to the reform.

The next three columns analyze the differential effect of the reform on female candidates for the three applicant characteristics used in the main tables: the age-adjusted standardized h index, whether the candidates received a PhD from the university to which they are applying, and whether the candidate's PhD supervisor sits in the committee. If anticipated discrimination plays a large role in women's decision to stay in academia and the gender quota modified these expectations, we would expect the relative quality of female candidates to change (Lepage et al. 2022). Homophily

Table 4. Effect of the Quota on Candidate Characteristics

	Dependent variable				
	Gender composition of candidate pool		Age-adjusted standardized h index	PhD from the same university	PhD supervisor in committee
	Estimator: OLS				
	(1)	(2)	(3)	(4)	(5)
Postquota $\times$ below 40% threshold	0.053 (0.05) [0.34]	−0.025 (0.05) [0.64]	—	—	—
Female candidate $\times$ postquota $\times$ below 40% threshold	—	—	−0.051 (0.29) [0.87]	−0.001 (0.07) [0.99]	−0.027 (0.06) [0.68]
Mean of dependent variable	0.39	0.39	0.00	0.06	0.24
Years	2009–2018	2013–2016	2009–2018	2009–2018	2009–2018
University $\times$ field fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Gender $\times$ university $\times$ field fixed effects	No	No	Yes	Yes	Yes
Gender $\times$ year fixed effects	No	No	Yes	Yes	Yes
Observations	455	296	1,357	1,357	1,357

Notes. Standard errors are clustered at the university $\times$ field level and can be found in parentheses. Wild-cluster bootstrapped p -values can be found in square brackets. The gender variable is coded as one for a woman and zero for a man. The treated group is constructed based on whether the average share of women in committees at the university-field level was below the 40% threshold prequota. Controls for columns (1) and (2) include president gender and professorship level. Controls for columns (3), (4), and (5) include a postquota $\times$ below 40% threshold dummy, president gender, professorship level, and the interaction of these last two variables with candidate gender.

in supervisor choice could also drive a differential effect on the number of connections that women have.

I estimate the following model using OLS:

$$Y_{ij} = \alpha_1 R_j T_j g_i + \alpha_2 R_j T_j + \beta_1 X_j + \beta_2 X_j g_i + \phi_j + \psi_j + g_i \phi_j + g_i \psi_j + \epsilon_{ij},$$

where Y_{ij} is the variable of interest. However, because the model is estimated at the candidate level, I interact all the fixed effects and independent variables with an indicator variable for the candidate's gender so that the reported coefficients can be interpreted as the differential effect of the quota for women relative to men.

The second row of Table 4 in columns (3), (4), and (5) reports α_1 , the effect of the quota on the three candidate characteristics (h index, PhD from the same university, and PhD supervisor in the committee, respectively) for women relative to men. The point estimates here are very small, and they suggest a negligible effect of the quota on these candidate characteristics for women relative to men.

Overall, although the estimation is imprecise, the results from this section suggest that the effect of the quota on supply-side characteristics is small relative to the effect on demand-side bias, at least in the short-term context of this study.

4.2. Changes in the Selection of Committee Members

In this subsection, I estimate the effect of the quota on the characteristics of committee members. The estimated models are all of the following type:

$$Y_j = \alpha R_j T_j + \gamma T_j + \delta R_j + \beta_1 X_j + \epsilon_j,$$

where Y_j is the variable of interest.

I first show the effect of the reform on the share of women among internal and external members in columns (1) and (2) of Table 5, respectively.¹⁶ The coefficients in the first row of Table 5 show that in order to meet the quota, committees increased the numbers of both external and internal female committee members, although they seem to draw more from their colleagues inside the department. This result implies that women inside the departments most affected by the reform have had to shoulder more administrative work post-2015 than their male colleagues, which could harm their ability to publish.

Did the reform affect the h index of committee members? We might surmise that the new female committee members have fewer publications than the male professors that they replace. In column (3) of Table 5, I show the effect of the quota on the average h index of committee members. There does not appear to be much of an effect of the reform, with the negative treatment effect on the mean h index of committee members being both small (approximately 0.1 of a standard deviation

in the h index) and insignificant. This shows that two concerns about the reform—that committee presidents choose junior women on purpose to negate the effects of the quota and that the new female committee members are less qualified than the members they replace, degrading the overall “quality” of the committee—do not seem to hold in the data.

In columns (4) and (5), I examine whether the way that committees are formed changed with the quota through two variables: the number of committee members and the probability that the president is female. Committee presidents are chosen by the university and not by members of the committee. For both variables, there is no significant effect of the quota. The lack of change in female presidents is consistent with the literature on boardroom quotas, which points out the lack of women in the positions of most power. For instance, Reberlioux and Roudaut (2019) find that, following the implementation of boardroom quotas in France, the newly promoted women are less likely to hold key positions in these boardrooms than newly promoted men. In columns (6) and (7), I study how the quota affects the selection of committee presidents. There is no detectable effect of the reform on the age or the h index of committee presidents, which along with the results in column (5), suggests that the reform does not affect the selection of committee presidents.

5. Discussion

5.1. Mechanisms

Thus far, the evidence in this paper cannot identify what is driving the worse outcomes for women after the reform, especially because most of the committee-level characteristics seem to have little to no effect. Women could prefer to hire men. This could be because they are expressing their true preferences or because women are penalized when they decide to promote women, as discussed in Hekman et al. (2017). Akerlof and Kranton (2000) postulate instead that men discriminate against women when their identities are threatened. It is possible that this is the case in our setting because having more women in a traditionally masculine setting could be seen as a threat to the masculinity of the committee members. The interpretation in this case would not be of opposite-gender bias in recruiting but of own-gender bias from men that appears when the gender identities of committee members are threatened.

A piece of evidence in favor of the latter hypothesis is presented in Table 6. In this table, the sample is split into committees headed by male and female presidents. I test the effect of the reform separately for these subsamples. The negative effect of the reform is driven entirely by committees with male presidents. Recall that Table 5 shows no difference in the likelihood of women being president or significant changes in the

Table 5. Effect of the Quota on Committee Characteristics

	Dependent variable						
	Share of women among internal members (1)	Share of women among external members (2)	Mean standardized h index of committee members (3)	Number of committee members (4)	Gender of committee president (5)	Age of committee president (6)	h index of committee president (7)
Estimator: OLS							
Postquota × below 40% threshold	0.176*** (0.04) [0.00]	0.105** (0.04) [0.03]	−0.095 (0.11) [0.41]	0.573 (0.41) [0.19]	−0.051 (0.09) [0.60]	1.061 (1.43) [0.49]	0.023 (0.19) [0.91]
Below 40% threshold	−0.171*** (0.03)	−0.183*** (0.04)	0.166 (0.10)	−0.276 (0.50)	−0.065 (0.11)	−1.017 (1.43)	−0.084 (0.18)
Postquota	−0.010 (0.03)	0.013 (0.03)	−0.024 (0.06)	−0.235 (0.21)	0.006 (0.07)	−0.227 (0.81)	−0.192 (0.14)
Mean of dependent variable	0.49	0.38	0.02	10.8	0.30	52.3	0.27
Field fixed effects	No	No	No	No	No	No	No
University fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	455	455	455	455	455	447	455

Notes. Standard errors are clustered at the university × field level and can be found in parentheses. Wild-cluster bootstrapped p -values can be found in square brackets. The gender variable is coded as one for a woman and zero for a man. The treated group is constructed based on whether the average share of women in committees at the university-field level was below the 40% threshold prequota. The h index is standardized by field and seniority.
** $p < 0.05$; *** $p < 0.01$.

Table 6. Effect of the Quota on the Hiring of Women Split by Gender of Committee President

	Dependent variable			
	Ranked first		Rank	
	Estimator: Conditional logit		Estimator: Rank-ordered logit	
	(1)	(2)	(3)	(4)
Interacted with gender of the candidate				
Postquota × below 40% threshold	3.288 (3.12)	−3.020** (1.28)	0.344 (0.73)	−1.393** (0.60)
Marginal effect of treatment	0.149	−0.279	0.0372	−0.155
Gender of president	Female	Male	Female	Male
Controls	All	All	All	All
Gender × university × field fixed effects	Yes	Yes	Yes	Yes
Gender × year fixed effects	Yes	Yes	Yes	Yes
Observations	391	963	391	963

Notes. Standard errors are clustered at the university × field level and can be found in parentheses. The gender variable is coded as one for a woman and zero for a man. The treated group is constructed based on whether the average share of women in committees at the university-field level was below the 40% threshold prequota. Contests where only one candidate was interviewed or where only applicants of a specific gender applied are dropped from the analysis. Controls include the share of female candidates, the standardized age-adjusted *h* index, president gender, professorship level, and academic connections. Marginal effects represent the average change in the probability of being ranked first for women in the treated group postquota.
***p* < 0.05.

age or *h* indices of committee presidents, so it is unlikely that these results are driven by changes in the selection of committee presidents following the reform. Results using committee member fixed effects are similar and are presented in Section 3 in the online appendix.

This result is consistent with the idea that men change their behavior as a result of the reform, but it leads to several different interpretations. These interpretations are not directly testable in the data but are documented in the literature, and they are not mutually exclusive. All could be relevant in this context.

The first possibility is that men are less likely to favor women once more women are on the committees, which corresponds well with the evidence seen in Figure 2. The graph shows a positive effect of being in the treated group prequota for women. A large body of literature in social psychology has discussed the existence of a moral licensing effect (Blanken et al. 2015), whereby previous good actions can make individuals more likely to commit immoral actions in the future. One possibility is, therefore, that the previously “good” moral actions of favoring women in the treated group when the committees were chosen with no quota constraints or of directly allowing more women into the committees led to a licensing effect where men felt more comfortable hiring other men following the implementation of the quota.

Licensing has been found to have an effect on hiring in controlled environments. For instance, individuals were more likely to express a preference for hiring White workers when given the opportunity to dismiss a racist action beforehand (Effron et al. 2012) or when writing about a Black friend (Bradley-Geist et al. 2010).

Monin and Miller (2001) show that this effect also holds with respect to gender in hiring. Picking a woman for a consulting job licensed participants to pick men for a more stereotypical male job.

Another possible explanation is that men become biased against women because of their opposition to the quota. Ip et al. (2020), for instance, show that gender quotas are more likely to be accepted when hiring discrimination against women is high but that quotas are generally seen as illegitimate, which leads to worse performance when there is no hiring discrimination. Leibbrandt et al. (2018) provide evidence from a laboratory experiment that women are more likely to be sabotaged when a gender quota is put in place. However, this effect is driven mostly by the differential reaction of women, not of men. These negative reactions to policies that promote gender equality have been termed the backlash effect in the literature and could be an explanation for the negative effect of gender quotas on the hiring of women found in this paper. It is important to note, however, that other papers in the field (Bagues et al. 2017) or in the laboratory (Domínguez 2023) find a negative reaction of men to the presence of women on committees, even in the absence of a gender quota. In Domínguez (2023), for instance, men are both more likely to be gender biased and to lead in female-dominated committees. Unless backlash is directed toward the mere presence of women on the committee, it seems unlikely that this is the only mechanism at play here.

A more subtle possibility is that women cannot express their true preferences when the committee president is male because doing so would lead to backlash

against them or because they want to fit in and avoid value threat (Duguid et al. 2012). Hekman et al. (2017), for instance, show that leaders from underrepresented groups receive worse evaluations if they advocate for candidates from their own group. This is consistent with previous research that shows that agentic women can face backlash (Rudman and Glick 2001). As a result, women have to develop strategies to both display competence and avoid backlash (Trzebiatowski et al. 2023). Because women can make a difference in the committee decision only if they are able to advocate, this could explain why the change in gender composition does not have a positive effect. A closely related explanation could be that men expect women to advocate for other women, and as a result, they adjust their own votes strategically. Snellman and Solal (2023) find that women-led firms that receive early funding from female-led venture capitalists are less likely to receive subsequent funding from other venture capitalists. They interpret this result as gender homophily signaling lower competence, a mechanism that could also operate in our setting.

Irrespective of the precise channel that causes this effect and consistent with the previous literature, the results presented in this article strongly suggest that gender quotas and other coercive measures may have unintended effects. In this specific case, increasing the proportion of female committee members led to a negative effect on the ranking of women by committees.

5.2. Limitations

One notable limitation of the paper is the lack of data on the CVs of candidates, which could provide valuable information on metrics other than publications that may have been taken into account by hiring committees. For example, committees with a higher proportion of women may place a greater emphasis on teaching or service, but I do not observe data on these dimensions. This is unlikely to change the main results, however, because women are typically more likely to take on nonpromotable tasks (Babcock et al. 2017), so we would expect this effect to improve women's probability of being hired relative to men, creating a downward bias in the estimation of the main effect.

Furthermore, although this study focuses on the impact of quotas on hiring committees and applicant characteristics, additional factors may affect our overall evaluation of the policy. For example, it is possible that new hires have research interests that align more closely with those of women on average, regardless of their gender. Moreover, hiring committees could offer networking opportunities for women, such as meeting new coauthors or journal editors. On the other hand, quotas could increase the administrative burden on women in academia. This burden is often not taken into account in discussions of tenure cases and could ultimately harm women's hiring and promotion prospects, particularly

in fields where they are already in the minority. Evaluating the effect of each of these channels, however, is beyond the scope of this paper.

Finally, this study does not directly observe the individual votes of committee members, which could provide more concrete evidence on whether it is the reaction of men that drives the effect of the reform. It would be of particular interest to combine individual reactions to the quota with a survey to test the different mechanisms that could explain the results.

5.3. External Validity

This paper provides evidence that gender quotas for hiring committees do not necessarily increase the share of women hired and in fact, can decrease it, adding to previous evidence suggesting that men are less favorable to women when randomly assigned to committees with more women (Bagues et al. 2017). Should we expect all efforts to increase diversity in hiring committees to have the same negative impact? Because the reaction of men drives the negative effect of the reform, it seems clear that designing an effective gender quota requires nullifying its impact on male behavior.

One promising design choice could be to decrease the influence of men in committees. Previous work has shown that although women are less influential than men in group decision making, this gender gap disappears when women are in the majority (Radbruch and Schiprowski 2023) and that female team leaders can help reduce discrimination against women (Stoddard et al. 2020). Imposing a 50% share of women in committees or accompanying the reform with a quota on female committee presidents could, therefore, yield more positive results than the quota in the current study. In a context where there are few women available to participate and participation is not necessarily rewarded in promotion decisions, this change in design might, however, prove particularly costly.

Another solution is to ensure that there is no negative reaction of men to the quota. It is complicated to know how to proceed given that we lack evidence on which precise psychological mechanism is at play, but one proposal could be to accompany the quota with efforts to change organizational culture (Luo and Zhang 2022). For instance, female representation has been shown to increase female hiring in similar committees (Baron et al. 2021) but only after a shift to a more supportive organizational culture triggered by interventions designed to raise participants' awareness of the benefits of gender diversity. Dobbin and Kalev (2018) suggests that organizations can avoid backlash against diversity initiatives by increasing manager engagement through, for instance, formal mentoring or management training programs. The hiring committees in this paper are created ad hoc for each separate recruitment, which could also be an important contextual element because previous research

has shown that individuals were less likely to hire women when making a single-hiring rather than multiple-hiring decision (Chang et al. 2020). Changing the hiring process so that a single committee handles all hires rather than many separate committees could potentially limit the unintended effects of the quota.

A second aspect is whether this paper is relevant for thinking about how similar reforms may play out in settings outside of academia. Although few other institutions have mandated quotas on the members of hiring committees, many organizations in both the public and private sectors face pressure to change their hiring practices. The implemented quota studied in this article has many similarities with board gender quotas, such as a similar 40% threshold and a lack of constraints on the most prestigious positions. Despite little evidence that board or hiring quotas trickle down, they remain popular with policy makers. For instance, a recent EU directive has imposed gender quotas on board membership for listed companies in EU member states. Lawmakers expect that “enhancing women’s participation in economic decision-making, on boards in particular, is expected to have a positive spill-over effect on women’s employment in the companies concerned and throughout the whole economy.”¹⁷ Given the evidence provided in this paper, institutions that want to improve their hiring of female candidates might want to consider other interventions in addition to diverse interview panels and gender quotas.

6. Conclusion

Are gender quotas in hiring committees a boon or a bane for women? In this article, I have presented evidence that should cast skepticism on the assumption that diversity on hiring committees could yield more diverse hires. In this setting, the reform had unintended negative effects, possibly caused by the reaction of men against the quota. This could be because of a licensing effect, backlash, or strategic adjustment. Although I cannot distinguish among these three mechanisms, my results suggest that any equalizing effects of the gender quota were more than offset by the negative demand-side response.

This paper has two major implications. The first relates to the unintended effects of gender quotas. Previous work has emphasized that diversity initiatives can have counterproductive effects because individuals resent constraints on their autonomy (Dobbin et al. 2015) and because they can create the illusion of a procedurally fair environment. The paper adds to this literature by showing that the behavior of men changes when more women are involved in the decision-making process, counterbalancing any effect of the increase in the number of women. To make progress on this issue, it is crucial to understand the reaction of men in particular

to these quotas, and thus, more work needs to be done to understand how gender affects group decision making (Born et al. 2020), especially the role of group leaders (Stoddard et al. 2020).

Second, the paper finds little effect of the quota on supply-side characteristics. Given that anticipated discrimination has been cited as an essential factor driving hiring inequalities (Pager and Pedulla 2015), a gender quota on the members of hiring committees could have altered beliefs and triggered a change in supply-side characteristics. Moreover, research has shown that employers signaling their value of diversity can affect the composition of the applicant pool (Kang et al. 2016). The reform has little effect on the composition of the applicant pool, the relative quality of female candidates, or their likelihood of being connected to someone in the committee, which suggests that the anticipated benefits of introducing gender quotas for supply-side characteristics are likely to be small. One possible explanation is that quotas imposed from the top down might do little to shift expectations or beliefs regarding discrimination. In contrast, voluntarily changing the composition of hiring committees could send a stronger signal about the organization’s culture and potentially shift beliefs about discrimination beyond the hiring stage (Solal and Snellman 2019).

The results from this paper have important practical consequences. They suggest that the underrepresentation of women might not be solved by increasing the share of women on hiring committees or interview panels, especially when taking the increased administrative work that women have to undertake into account. These results should provoke caution in decision makers wanting to promote new diversity initiatives and consideration of whether these policies can have unintended consequences.

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Appendix A. Main Robustness Checks

A.1. Shorter Time Frame

One potential concern is the relatively long time span in my data, which could lead to confusing long-run differences between the treated and comparison groups with the effect of the reform. Table A.1 presents results using the years immediately leading up to and following the reform to alleviate this concern. Columns (1) and (2) present the results based on this limited time frame, including the years from 2013 to 2016. The point estimates are once again very close to those presented in the other tables, so we can probably rule out the idea that the estimated effect of the reform is because of longer-run trends.

A.2. Top Ranks Only

One concern with using ranked data is that some committees may not care about how they rank lower-quality candidates because they are very unlikely to be hired. Our results could thus be driven by inattention from committee members toward candidates less likely to be recruited rather than discrimination as in Bartoš et al. (2016). To investigate this concern, I present results in columns (3) and (4) of Table A.1 again using the conditional and rank-ordered logit specifications, but I keep only the top three candidates from each

committee and drop candidates who were interviewed but were not ranked. I keep the two years on either side of the reform only as in columns (1) and (2). In this specification, the reform still has a significant effect, and the point estimates are even larger than those that consider the full ranking. We can, therefore, rule out that the effect of the reform is driven by inattention or lack of effort from committees in ranking candidates who are very unlikely to be hired.

A.3. Continuous Specification

When using a binary variable to compute the effect of the reform, fields far away from the cutoff before the reform are weighted in the same away as those that were close. Another way to proceed is to use the distance of the average share of women in committees prequota to the 40% threshold as a treatment variable instead. I use the following specification to estimate the effect of the reform on the ranks of women:

$$Y_{ij} = \mu_{ij}g_iR_j\max\{0, 0.4 - \text{committee}_{ij}^{pre-2015}\} + \beta_1X_i + \beta_2g_iX_j + g_i\phi_j + g_i\psi_j + \epsilon_{ij},$$

where ϕ and ψ are year and university $\times$ field fixed effects, committee is the mean share of female committee members at the field-university level before 2015, and Y is the dependent

Table A.1. Effect of the Quota on the Rank of Female Candidates—Limited Years, Top Three Candidates, and Continuous Specification

	Dependent variable					
	Ranked first Estimator: Conditional logit (1)	Rank Estimator: Rank-ordered logit (2)	Ranked first Estimator: Conditional logit (3)	Rank Estimator: Rank-ordered logit (4)	Ranked first Estimator: Conditional logit (5)	Rank Estimator: Rank-ordered logit (6)
Interacted with gender of the candidate						
Postquota $\times$ below 40% threshold	−2.267** (1.13)	−0.863*** (0.33)	−2.729** (1.14)	−1.812** (0.73)		
Distance to threshold $\times$ postquota					−18.410** (9.24)	−7.595** (3.52)
Variables at the candidate level						
Standardized age-adjusted h index	0.452*** (0.09)	0.318*** (0.06)	0.389*** (0.11)	0.287*** (0.10)	0.354*** (0.08)	0.286*** (0.05)
Marginal effect of treatment	−0.207	−0.104	−0.315	−0.283	−0.184	−0.0857
Years	2013–2016	2013–2016	2013–2016	2013–2016	2009–2018	2009–2018
Ranks	All	All	Top 3	Top 3	All	All
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Gender $\times$ university $\times$ field fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Gender $\times$ year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	926	926	597	597	1,357	1,357

Notes. Standard errors are clustered at the university $\times$ field level and can be found in parentheses. The gender variable is coded as one for a woman and zero for a man. The treated group is constructed based on whether the average share of women in committees at the university-field level was below the 40% threshold prequota. Contests where only one candidate was interviewed or where only applicants of a specific gender applied are dropped from the analysis. Controls include the share of female candidates, the standardized age-adjusted h index, president gender, professorship level, and academic connections. Variables interacted with the gender of the candidate should be interpreted as the differential effect of the variable for women with respect to men. Only the two years on either side of the reform are included in columns (1)–(4). Only the top three candidates are considered in columns (3) and (4). The marginal effects in columns (1)–(4) represent the average change in the probability of being ranked first for women in the treated group postquota. The marginal effects in columns (5) and (6) show the difference in the predicted probability of being ranked first for treated women postquota following a 10% increase in the distance to threshold.

** $p < 0.05$; *** $p < 0.01$.

Table B.1. Gendered Effects of Networks

	Dependent variable: Rank			
	Estimator: Rank-ordered logit			
	(1)	(2)	(3)	(4)
Variable at the candidate level				
PhD supervisor in the committee	0.736*** (0.22)	0.758*** (0.22)		
PhD supervisor in the committee × candidate is female	−0.517** (0.23)	−0.564** (0.25)		
PhD from the same university			0.356*** (0.11)	0.376*** (0.12)
PhD from the same university × candidate is female			−0.060 (0.20)	−0.091 (0.20)
Standardized age-adjusted <i>h</i> index	0.274*** (0.03)	0.279*** (0.03)	0.279*** (0.03)	0.283*** (0.03)
Candidate is a woman	−0.191*** (0.07)	−0.132 (0.24)	−0.216*** (0.07)	−0.145 (0.24)
Gender × field fixed effects	No	Yes	No	Yes
Observations	1,357	1,357	1,357	1,357

Notes. Standard errors are clustered at the university × field level and can be found in parentheses. The gender variable is coded as one for a woman and zero for a man. Contests where only one candidate was ranked or where only applicants of a specific gender applied are dropped from the analysis.

p* < 0.05; *p* < 0.01.

variable of interest: the rank or whether the candidate finished first depending on the specification. This specification is reminiscent of the specification used in Bertrand (2011), the main difference being that there are multiple observations by treatment unit. Our parameter of interest here is μ_{ij} , which should capture the effect of the quota. The results for the conditional logit and the rank-ordered logit can be seen in columns (5) and (6) of Table A.1, and they are similar to those presented in the previous tables. The results remain quantitatively similar when I use distance to the threshold on the other side rather than normalizing all departments with female shares in committee above 40% before the reform to zero. The marginal effects presented here show the difference in probability of being ranked first for treated women postreform following a 10% increase in the distance to the threshold.

Appendix B. Differentiated Effects of Networks

Qualitative research has shown that women may benefit less from academic connections than men do and that thesis supervisors are more likely to abstain during committee discussions if their PhD student is a woman (see the section by Sautier in Deschamps et al. 2020). In my data set, women are more likely to have their supervisor on the committee than men (48 women and 42 men), but the gains from this connection may not be the same for both genders. I directly test this hypothesis in Table B.1 by interacting the effect of having one’s advisor on the committee with the gender of the candidate using the rank-ordered logit framework. The results are striking; women benefit much less from having their supervisor on the committee than men do. In fact, the point estimates imply that the positive effect of having one’s thesis supervisor in the committee for a woman is almost exactly canceled out by the negative effect of being a woman. In other words,

having one’s supervisor in the committee enables women to be ranked the same as men. I test for gender differences in the effect of doctoral networks in columns (3) and (4). There are no significant or large differences in the effect of having received a PhD from the institution where one is applying.

Endnotes

¹ Academia is not exempt from the stark reality of gender gaps depicted. Even though women make up the majority of PhD candidates in many academic fields, they are underrepresented in faculty positions, and the gender gap increases with seniority. In the European Union (EU), for instance, women made up 47% of PhD graduates and 40% of assistant professors, but they made up only 23% of full professors, according to the European Commission’s 2018 She figures. France, our country of interest, has a lower share of female researchers than the EU average. Beyond this vertical segregation, academia also suffers from horizontal segregation, with women much less likely to be studying and working in Science, Technology, Engineering and Mathematics (STEM).

² Paola and Scoppa (2015) find positive effects on the hiring of women but only when the number of women in the committee increases from zero to one.

³ See, for example, for employment at the Federal Reserve (<https://oig.federalreserve.gov/reports/bureau-hiring-practices-diverse-workforce-mar2021.pdf>) and in the Australian (<https://www.vic.gov.au/node/24093>) and UK governments (https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/800602/Research-on-Diverse-SCS-Interview-Panels-Memorandum-of-Understanding.pdf).

⁴ Similar arguments for diverse interview panels are made by PwC at <https://www.pwc.com/gx/en/about/diversity/iwd/iwd-female-talent-report-web.pdf>.

⁵ A scientist’s *h* index is the *h* number of publications he has over *h* citations. I also collect citation and *h* indices, which are discounted

by the number of coauthors and how long ago the article was published. Descriptive statistics on the h indices of candidates and the other candidate characteristics are presented in the online appendix.

⁶ In fields with very few women, such as political science, mathematics, and engineering, the ratio for parity was lower and set each year by decree. This is why the Maths 2 Department is to the left of the 40% threshold after the reform. In other fields, this ratio is binding, but once the committee has been approved, if a committee member drops out, the replacements do not need to respect the gender quota. The data that I have reflect only the final composition of the committee.

⁷ I directly estimate the link between the share of women on the committees and the hiring of women in Section 2 in the online appendix. However, these are only correlational results. I rely instead on the quota to generate variation in the gender composition of these committees.

⁸ Many studies examining gender bias in hiring conceptualize the hiring process as composed of a series of sequential steps (application-interview-offer). The quota could in theory affect the step between the application and interview stage, but I have no information on who the noninterviewed applicants are, which makes a detailed study of this step impossible. Nevertheless, I study how the quota affects the difference between the gender composition of interviewed candidates and applicants in Table OA.44 in the online appendix.

⁹ The standard errors presented in the paper are clustered at university-field level. Cameron et al. (2008) document that these standard errors can be biased when the number of clusters is below 40, as is the case in this paper. They recommend the use of wild-cluster bootstrapping to obtain the correct p -values, a method that is available only for methods with linearly additive errors, such as OLS, and not for the conditional and rank-ordered logits. When I use this bootstrapping procedure for the OLS results, the p -values for the effect of the quota move from 0.040 to 0.041 in column (1) and from 0.031 to 0.037 for column (2). This suggests that there is little evidence of an overrejection problem in the data and that the bias from the small number of clusters in the nonlinear estimations is likely to be small.

¹⁰ This approach can be rationalized by considering that the committee has a latent utility function and ranks the candidates based on this function, the form of which is given by the estimated specification. Using an OLS regression to estimate the probability of being first ranked on the candidate data, on the other hand, is problematic because the observations are not independent. Within the committee, if candidate A is hired, then candidate B cannot be.

¹¹ This is Maitre de Conférences or Professeur des Universités in French. Both are included in the analysis. Table OA.26 in the online appendix shows that there is no differential effect of the reform when comparing these two types of positions separately.

¹² The coefficients shown in the tables are those from the latent committee utility function and not the marginal effects. Ai and Norton (2003) warns that the marginal effects of interaction terms may not be correctly estimated in nonlinear models and may even have the wrong sign depending on the support. The treatment effect presented here, however, is estimated as a nonlinear difference in differences, which Puhani (2012) shows is correctly estimated by the interaction effect in this specific case. Nevertheless, I show that the interacted effect estimated using the Ai and Norton (2003) method does not flip over the support.

¹³ I use the age-adjusted h index of candidates instead of citations because this is the publication measure that is the most significant when including all other measures, as seen in Table OA.21 in the online appendix. This measure is also standardized by field and seniority.

¹⁴ I compute this marginal effect by first using the estimated coefficients to recover the probability of being ranked first. I then re-estimate the predicted probabilities when setting the effect of the quota equal to zero. Next, I compute the difference between these two probabilities for women in the treated group, which corresponds to the effect of the quota on the probability of being first for these women. The estimated figure is then the average difference in probabilities for women affected by the quota after the reform.

¹⁵ More details on how we can use the rank-ordered logit to estimate gender bias can be found in Section 4 in the online appendix. Results from simulations in Section 6 in the online appendix show that this is also the best method to reliably detect discrimination given the data-generating process in the model section.

¹⁶ In order to show the sizes of γ and δ , I only include university fixed effects. As shown in the online appendix, the results are quantitatively similar when the full set of fixed effects is added.

¹⁷ See <https://eur-lex.europa.eu/eli/dir/2022/2381/oj>.

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